Publication list of Giordano De Marzo

Publications

Publications marked with ⋆ are the three most important papers, submitted with this application. Google Scholar (October 2026): h-index 15, i10-index 21, 718 citations. ORCID 0000-0002-3127-5336.

[P1] ⋆ De Marzo, Giordano, Claudio Castellano, and David Garcia. “AI agents can coordinate via majority-following beyond human scale.” Science Advances 12, eaea6091 (2026).

[P2] Bouleimen, Azza, Giordano De Marzo, Taehee Kim, Nicolò Pagan, Hannah Metzler, Silvia Giordano, and David Garcia. "The Collective Turing Test: Large Language Models Can Generate Realistic Multi-User Discussions." Scientific Reports (2026).

[P3] Calò, Emanuele, Giordano De Marzo, and Vito DP Servedio. "Species vulnerability and ecosystem fragility: A dual perspective in food webs." Chaos, Solitons & Fractals 199 (2025): 116741.

[P4] Bellina, Alessandro, Giordano De Marzo, and Vittorio Loreto. "Modelling singularities in macroevolution." European Economic Review 177 (2025): 105052.

[P5] Sylos Labini, Francesco, Giordano De Marzo, and Matteo Straccamore. "Breaking the degeneracy between warps and radial flows in external galaxies." The Astrophysical Journal 988, no. 1 (2025): 122.

[P6] Bellina, Alessandro, Giordano De Marzo, and Vittorio Loreto. "Full spectrum of novelties in time-dependent urn models." Physical Review Research 7, no. 2 (2025): 023127.

[P7] Servedio, Vito DP, Alessandro Bellina, Emanuele Calò, and Giordano De Marzo. "Fitness centrality: a non-linear centrality measure for complex networks." J. Phys. Complex. 6, 015002 (2025).

[P8] Labini, Francesco Sylos, Roberto Capuzzo-Dolcetta, Giordano De Marzo, and Matteo Straccamore. "Exploring the dark matter disk model in dwarf galaxies: Insights from the LITTLE THINGS sample." Astronomy & Astrophysics 693 (2025): A248.

[P9] Di Bona, Gabriele, Alessandro Bellina, Giordano De Marzo, Angelo Petralia, Iacopo Iacopini, and Vito Latora. "The dynamics of higher-order novelties." Nature Communications 16.1 (2025): 393.

[P10] De Marzo, Giordano, Pietro Gravino, and Vittorio Loreto. "Recommender systems may enhance the discovery of novelties." Journal of Physics: Complexity 5, no. 4 (2024): 045008.

[P11] Aufiero, Sabrina, Giordano De Marzo, Angelica Sbardella, and Andrea Zaccaria. "Mapping job fitness and skill coherence into wages: an economic complexity analysis." Scientific Reports 14, no. 1 (2024): 11752.

[P12] Sylos Labini, Francesco, Giordano De Marzo, Matteo Straccamore, and Sébastien Comerón. "The Tully–Fisher relation and the Bosma effect." Monthly Notices of the Royal Astronomical Society 527, no. 2 (2024): 2697-2717.

[P13] ⋆ Bellina, Alessandro, Claudio Castellano, Paul Pineau, Giulio Iannelli and Giordano De Marzo. "Effect of collaborative-filtering-based recommendation algorithms on opinion polarization." Physical Review E 108.5 (2023): 054304. Contribution (approx. 40%): conceived the study and the model, supervised the junior researchers who carried out the work (a former master’s student of mine and a research intern), and led the writing of the manuscript; senior author.

[P14] Sylos Labini, Francesco, Matteo Straccamore, Giordano De Marzo, and Sébastien Comerón. "Mapping non-axisymmetric velocity fields of external galaxies." Monthly Notices of the Royal Astronomical Society 524, no. 1 (2023): 1560-1600.

[P15] De Marzo, Giordano, and Vito DP Servedio. "Quantifying the complexity and similarity of chess openings using online chess community data." Scientific Reports 13.1 (2023): 1-12.

[P16] De Marzo, Giordano, Federico Attili, and Luciano Pietronero. "Growing inequality in systems showing Zipf's law." Journal of Physics: Complexity 4.1 (2023): 015014.

[P17] De Marzo, Giordano, and Giulio Iannelli. "Effect of spatial correlations on Hopfield Neural Network and Dense Associative Memories." Physica A: Statistical Mechanics and its Applications 612 (2023): 128487.

[P18] De Marzo, Giordano, Francesco Pandolfelli, and Vito DP Servedio. "Modeling innovation in the cryptocurrency ecosystem." Scientific Reports 12.1 (2022): 12942.

[P19] De Marzo, Giordano, et al. "Quantifying the unexpected: A scientific approach to Black Swans." Physical Review Research 4.3 (2022): 033079.

[P20] Iannelli, Giulio, Giordano De Marzo, and Claudio Castellano. "Filter bubble effect in the multistate voter model." Chaos: An Interdisciplinary Journal of Nonlinear Science 32.4 (2022): 043103.

[P21] Agliari, Elena, et al. "The emergence of a concept in shallow neural networks." Neural Networks 148 (2022): 232-253.

[P22] De Marzo, Giordano, Francesco Sylos Labini, and Luciano Pietronero. "Zipf's law for cosmic structures: How large are the greatest structures in the universe?." Astronomy and Astrophysics 651 (2021): A114.

[P23] De Marzo, Giordano, et al. "Dynamical approach to Zipf's law." Physical Review Research 3.1 (2021): 013084.

[P24] Agliari, Elena, and Giordano De Marzo. "Tolerance versus synaptic noise in dense associative memories." The European Physical Journal Plus 135.11 (2020): 1-22.

[P25] ⋆ De Marzo, Giordano, Andrea Zaccaria, and Claudio Castellano. "Emergence of polarization in a voter model with personalized information." Physical Review Research 2.4 (2020): 043117.

Preprints

[W1] De Marzo, Giordano, Nicolò Alboré, and David Garcia. "Copying explains the collective behavior of AI agents in the wild." arXiv preprint arXiv:2609.09150 (2026).

[W2] Keßler, Raphaela, R. D. Ventzke, Viola Priesemann, and Giordano De Marzo. "Network Information Enhances Unreliable News Domain Detection." arXiv preprint arXiv:2608.02399 (2026).

[W3] De Marzo, Giordano, Alessandro Bellina, Claudio Castellano, Viola Priesemann, and David Garcia. "Conformity Generates Collective Misalignment in AI Agents Societies." arXiv preprint arXiv:2605.10721 (2026). In revision at Nature Communications.

[W4] Fenoaltea, Enrico Maria, Francesco Santoro, Giordano De Marzo, Segun Taofeek Aroyehun, and Andrea Tacchella. "Anticipating Innovation Using Large Language Models." arXiv preprint arXiv:2605.04875 (2026).

[W5] Bellina, Alessandro, Gabriele Di Bona, Giordano De Marzo, and Vittorio Loreto. "The gold-rush effect: how innovation speeds up." arXiv preprint arXiv:2602.13775 (2026).

[W6] De Marzo, Giordano, and David Garcia. "Collective Behavior of AI Agents: the Case of Moltbook." arXiv preprint arXiv:2602.09270 (2026).

[W7] Bellina, Alessandro, Giordano De Marzo, and David Garcia. "Conformity and Social Impact on AI Agents." arXiv preprint arXiv:2601.05384 (2026).

[W8] Marydas, Siddharth, Nanditha Mathew, Giordano De Marzo, and Carlo Pietrobelli. "Digital Technologies, Hiring, Training, and Firm Outcomes." MERIT Working Paper (2025).

[W9] Servedio, Vito DP, Alessandro Bellina, Emanuele Calò, and Giordano De Marzo. "Economic Complexity in Mono-Partite Networks." arXiv preprint arXiv:2405.04158 (2024).

[W10] De Marzo, Giordano, Luciano Pietronero, and David Garcia. "Emergence of scale-free networks in social interactions among large language models." arXiv preprint arXiv:2312.06619 (2023).

[W11] De Marzo, Giordano, Nanditha Mathew, and Angelica Sbardella. "Who creates jobs with broad skillsets? The crucial role of firms." ILO Working Paper No. 94 (2023).